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SUMMARY

ML systems architect, close to a decade across enterprise banks, an AI research lab and AI-native startups. I build and operate **Striff**, which checks every pull request against the architecture documents already in the repository: a language model turns the documents' sentences into rules, and a program decides whether the code still satisfies them. Before that: ML and GenAI platforms from ingestion through serving and monitoring on AWS, GCP and Kubernetes; a hybrid Kubernetes migration at **RBC Royal Bank** under bank change management; ML research infrastructure at **BorealisAI** on NVIDIA DGX clusters.

Core Capabilities

Production · public reference implementations

LLM & GenAI Systems · RAG · Embeddings

- › Production vLLM serving on Kubernetes via KubeAI with scale-from-zero, KV-cache routing and OpenAI-compatible APIs. For Joinable this removed the dependence on external inference APIs ([self-hosted LLM stack write-up](#)).
- › Embedding retrieval, RAG with traceable outputs, and self-hosted models (DeepSeek, Llama 3) beside frontier APIs (GPT, Claude, Bedrock).

[hadi-technology/vllm-mlops](#)

Verified LLM Output · Graph RAG · NLP

- › Systems where the model proposes and a program verifies: Striff's documented rules are translated by a model, then type-checked and evaluated deterministically against the parsed code, so no finding exists that a program did not compute.
- › Graph RAG with ShEx-constrained schema validation and a validator-LLM feedback loop for accuracy and traceability where pure vector RAG fails ([Fylo pipeline write-up](#)). Transformer NLP/NER pipelines for entity and sentiment extraction at production scale.

[hadi-technology/striff-lib](#) · [hadi-technology/clarpse](#)

End-to-End MLOps · CI/CD/CT · Monitoring

- › Pipelines with MLflow tracking, gated promotion, Argo CD GitOps and blue/green model rollouts via Argo Rollouts. Promotion is a merge, not a redeploy ([reference architecture](#)).
- › Drift detection, model versioning, automated retraining hooks and Vault-backed governance; for BrandWell, reproducible MLflow workflows with gated model promotion.

[hadi-technology/mlops-blueprint](#)

Cloud-Native Platforms · K8s · Terraform · GPU

- › Multi-cloud ML platforms across AWS, GCP and on-prem Kubernetes; hybrid container infrastructure under bank change management; Terraform-driven IaC. For SkanAI, model deployment time came down to minutes.
- › GPU orchestration with autoscaling and continuous batching; observability with Prometheus, Grafana and EFK; compliance-aligned for OSFI, change advisory boards and audit-grade deployment.

RBC Royal Bank · 2021-2022 · OSFI-aligned · **BorealisAI** · NVIDIA DGX

PRODUCTION-GRADE

ML systems serving live workloads, not proofs of concept. Async pipelines, autoscaled inference, blue/green rollouts under sustained load.

ENTERPRISE CONTEXT

Built under bank change management at **RBC**; ML research compute at scale at **BorealisAI** on NVIDIA DGX clusters with SLURM orchestration.

TECHNICAL LEADERSHIP

Founder and lead engineer at HADI Technology. Lead architect across 7+ client engagements. Peer-reviewed research at an IEEE-affiliated venue.

Reference Project

Live · public · end-to-end

striff.io → striff.io

[GitHub App](#) [live](#) [open-core](#)

Striff reads the documents already in a repository, READMEs, architecture notes and agent instruction files such as AGENTS.md, and checks every pull request against them. A language model translates each sentence that makes a claim about the code into a rule in a small formal language; a program evaluates the rule

STACK

- › Java · Python
- › MongoDB · Redis
- › Kubernetes · Terraform
- › Argo CD
- › Prometheus · Grafana

against the parsed code at both revisions of the pull request and posts the verdict as a GitHub check run beside the sentence it came from.

What makes it hold up. Source is parsed without compiling, so a claim the parser cannot answer is reported as unanswerable, never as a pass. Rules naming things the code model cannot hold are refused before evaluation. Every finding cites the document line and the code witness. Spring Boot on Kubernetes, MongoDB as the system of record, a Redis queue with priority lanes, Argo CD delivery.

The parser builds on research published at **ICCQ 2021 in cooperation with IEEE Computer Society**. striff-lib is open source on Maven Central as `io.github.hadi-technology:striff-lib`. Architecture write-up at mfadhel.com.

PUBLIC ARTIFACTS

[hadi-technology/striff-lib](https://github.com/hadi-technology/striff-lib)
[hadi-technology/clarpse](https://github.com/hadi-technology/clarpse)
mfadhel.com/striff-io-ml-infrastructure

Engagement Modes

Three ways to start

01 · Assess

Platform Review

Current state, target architecture, GPU economics, gaps. Written report and a working session.

3–4 weeks · fixed-price

02 · Build

Implementation

Self-hosted LLM serving, ML platforms, async pipelines, observability, governance.

8–16 weeks · outcome-priced

03 · Operate

Managed Platform

Ongoing operations, model lifecycle, GPU cost optimization, on-call coverage.

monthly retainer

Track Record

Verifiable engagements

2021 – 2022 · Contract

RBC Royal Bank *Cloud Infra Eng.*

Migration of a critical Apigee Edge platform to hybrid AKS + Anthos under bank change management. GitOps with Flux/Kustomize, an automation framework for cluster operations and credential rotation, chaos testing of runtime services.

Apigee · AKS · Anthos · Flux · Ansible · OSFI

2019 – 2020 · Full-time

BorealisAI *ML Infra Dev.*

RBC's AI research lab. SLURM on NVIDIA DGX clusters for research compute. PureStorage S3 adoption with a 100%+ ingestion throughput gain. OpenShift CI/CD for ML deployments with Jenkins, Vault and Artifactory.

SLURM · NVIDIA DGX · OpenShift · Jenkins · Vault · Artifactory

2018 – 2019 · Full-time

Buzz Indexes *MLOps Eng.*

NLP extraction pipelines turning unstructured financial text into structured entity and sentiment signals powering live equity indexes. Migrated to event-driven serverless on AWS, cutting pipeline runtime by 50%.

AWS · ECS · Lambda · NLP · entity extraction

Recent Client Engagements

via HADI Technology · 2023+

SkanAI ↗

Built ML inference infrastructure for financial services intelligence; reduced model deployment time to minutes.

Joinable ↗

Designed and deployed self-hosted LLM serving on Kubernetes; removed the dependence on external inference APIs.

FyloAI ↗

Built the scientific graph RAG and extraction pipeline with ShEx-constrained LLM loops.

ZahraTrust ↗

Architected AI agent infrastructure with traceable inference and controlled tool-use boundaries.

BrandWell ↗

Operationalized content ML systems with reproducible MLflow workflows and gated model promotion.

Moxby

Stood up the ML platform foundation: cross-cloud pipelines, artifact versioning, observability stack.

Terradotta ↗

Designed ML pipeline operations with drift monitoring and structured human-in-the-loop validation.

Contact mfadhel.com/contact

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